The Physics of the Instant
Analysis from Limits, Relativity, and Renormalization*

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Abstract

Much of theoretical physics rests on the idealization that a physical system possesses a complete state at each instant of time, a state sufficient to determine its future evolution. We argue that this completeness is a physical assumption rather than a consequence of how the instantaneous state is defined, and that established physics provides no basis for it and positive reasons to doubt it. Three independent lines of evidence are assembled. First, operationally the state at a temporal point is the continuous extension of descriptions over finite intervals: the ε-δ definition of a limit assigns a value at the point from its punctured neighborhoods, and whether that value is dynamically sufficient is a property of the dynamics that the construction neither supplies nor guarantees. Second, quantum mechanics and general relativity, applied jointly, bound spatial and temporal resolution at the Planck scale, and the Salecker-Wigner clock bound, combined with the requirement that the clock itself not undergo gravitational collapse, yields the same temporal bound from the properties of the clock rather than of the probed region. These bounds are not theorems of a completed theory of quantum gravity, but within the semiclassical domain in which they are derived they obstruct the limiting process $\delta t \to 0$. Third, in quantum field theory products of field operators at coincident points are singular, an effective description carries a validity scale beyond which it fails, Wilson's analysis makes renormalization a feature of the theory rather than a defect of perturbation theory, with a continuum limit requiring an ultraviolet fixed point of the flow, and the algebraic formulation assigns observables to open regions rather than to points and fixes the dynamics from any neighborhood of a Cauchy surface while assigning no algebra to the surface itself. Taken together, these show that completeness at a temporal point is not delivered by the construction of the state, is not reachable by measurement within the domain of established physics, and is not where established theories place their physical content. Dynamical completeness is defined for the deterministic and the stochastic settings, and the descriptions that presuppose it, the uniquely solvable initial-value problem and the Markov property, inherit the assumption. For an irreducible finite-state Markov chain in its stationary law, the composition defect of its temporal window average vanishes at least linearly with the window width, which yields a one-sided test of memorylessness on the observed state space. The memoryless description is the special case that requires justification.

Keywords: instantaneous state; dynamical completeness; Markov property; temporal resolution; Planck time; Salecker-Wigner clock bound; renormalization group; effective field theory; time-slice axiom

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1 Introduction

The assumption that a physical system possesses a complete state at each instant of time is so deeply embedded in theoretical physics that it is rarely stated as an assumption at all. In classical mechanics, the state is specified by positions and momenta at a time t; in quantum mechanics, by a vector in Hilbert space or equivalently a density matrix; in classical field theory, by field configurations and their conjugate momenta on a Cauchy surface. In every case, the formalism presupposes that the state at a temporal point carries the full kinematic specification of the system: all degrees of freedom, all constraints, all information required to determine the system's future evolution.

This paper argues that completeness is a separate physical assumption, not a property the instantaneous state inherits from the way it is constructed, and that established physics provides no basis for it and supplies positive reasons for doubt. Three independent arguments support this. One, from the mathematics of limits, shows that the construction defining the instantaneous state delivers an element of the state space at the point and nothing about its dynamical sufficiency, which is a property of the dynamics that the construction does not address. The second, from relativistic gravitation and quantum clock bounds, shows that within the semiclassical domain of those bounds the physical process of reaching the temporal point is obstructed at the Planck scale, so that completeness at the point is an assumption that no measurement within that domain can underwrite. The third, from quantum field theory, shows that the established formalism becomes singular exactly at the point-valued limit and that physical content is located in extended regions. The claim is that the completeness of an instant must be argued for rather than presupposed, not that an instant is empty, and that the argument runs opposite to the usual direction: the memoryless description, in which the state at each time suffices for all later times, is the case requiring justification.

The assumption has a precise form in each of the two settings in which it arises.

Definition 1 (Dynamical completeness)**.** (a) In a deterministic theory with state space S, the instantaneous state is dynamically complete if the initial-value problem is uniquely solvable from data on a single time slice: for every admissible $x(t_0) \in S$ there is a unique solution on $[t_0, \infty)$, determined by $x(t_0)$ alone. (b) In a stochastic theory, a process (X_t) with values in S is dynamically complete at instants if it has the Markov property with respect to its natural filtration $(\mathcal{F}_s)$: $\mathbb{P}(X_t \in A \mid \mathcal{F}_s) = \mathbb{P}(X_t \in A \mid X_s)$ for all $s < t$ and all measurable $A \subseteq S$.

In sense (b) the Markov property implies that the two-time transition kernels $\Gamma(t \leftarrow s)(A \mid x) = \mathbb{P}(X_t \in A \mid X_s = x)$ compose through every intermediate time, which is the Chapman-Kolmogorov relation written out in Sec. 6. The converse fails: Feller [4] exhibited a non-Markov process whose two-time kernels compose. Composition is therefore a necessary condition for completeness in sense (b), so that its failure refutes completeness while its holding does not establish it. The initial-value formulation of dynamics presupposes completeness in sense (a); the Markov property is completeness in sense (b); the notion of an instantaneous collapse in quantum measurement presupposes it for the quantum state. Sections 2 to 4 concern the point-indexed state as such and bear on both senses; Sec. 5 concerns the quantum state, for which the presupposition takes sense (a) in the Hilbert-space formulation and sense (b) in the stochastic formulation of the correspondence discussed there; Sec. 6 concerns sense (b) on a specified observed state space, where data can test composition. In each case the presupposition concerns the completeness of the state at a temporal point in the relevant sense, not its existence, which is a matter of definition and is not in dispute here. This paper examines that presupposition directly. Each of the three arguments below draws on established physics; the new element is their joint bearing on whether completeness at a temporal point is delivered by anything in physics. Section 6 draws the consequences for memoryless descriptions and the connection to

the indivisible stochastic processes of Barandes [1, 2, 3].

The state at a temporal point can be defined, and nothing here disputes the definition. The question is whether the object so defined inherits the property dynamical frameworks demand of it, the complete kinematic information sufficient to fix future evolution. The construction does not supply that property, and established physics provides independent reasons to doubt it.

2 Argument from Limits

Theories introduce point-indexed states directly: a trajectory $x(t)$, a density operator $\rho(t)$, Cauchy data on a slice. The state of a physical system at a temporal point is nonetheless not directly observable [5]: every physical measurement ever performed has yielded an average over a nonzero interval, and the point-indexed object is confronted with experiment only through such averages. Operationally, then, the state at time t is the limit of interval averages under a stated observation model, and the model used here is the uniform average over a window ending at t: for a state-valued function $s \mapsto \rho(s)$ write

$$\bar{\rho}_\delta(t) = \frac{1}{\delta} \int_{t-\delta}^{t} \rho(s)\, ds; \tag{1}$$

if ρ is continuous in the topology of the state space, then $\bar{\rho}_\delta(t) \to \rho(t)$ as $\delta \to 0$. In standard analysis the infinitesimal interval has no existence; one has either finite intervals or the point, with nothing in between. The question is therefore what the point inherits from the intervals that define it operationally, and in particular whether the completeness attributed to the point-indexed object has any warrant beyond stipulation. The conclusion of this section already follows from the meaning of dynamical sufficiency in Definition 1, since sufficiency is a property of the dynamics and not of any construction of the state; the construction is examined because it shows where the stipulation enters.

The ε-δ definition of the limit is mathematically rigorous [6], and the limit value exists and is well defined. The question is whether the limit point inherits the properties that dynamical frameworks demand of it. For a function f and a point a, the statement $\lim_{x\to a} f(x) = L$ means: for every $\varepsilon > 0$, there exists $\delta > 0$ such that for all x with $0 < |x - a| < \delta$, we have $|f(x) - L| < \varepsilon$. The condition $0 < |x - a|$ excludes the point a itself: the limit characterizes the behavior of f in every punctured neighborhood of a while making no assertion about a. The exclusion makes the limit independent of any value separately assigned at the point, and it is the reason limits are useful: they allow a value to be assigned where the function is not defined, or where its value differs from the limiting behavior. The limit of $\sin(x)/x$ as $x \to 0$ is 1, although the function is undefined at $x = 0$; the continuous extension assigns the value there by stipulation. The construction delivers the value and no other property of the function at the point.

The limit in (1) is taken in the state space, a phase space or a space of density operators, and is an element of it with all the structure that membership confers. The construction cannot supply the further property that this element determines the future, because sufficiency is a property of the dynamics, the existence of a law that fixes later states from the present one alone, and not a property of the limit. Two stipulations therefore enter wherever a point-indexed complete state is written down. The first is continuity in time, which identifies the value at the point with the limit of the approach; it is a physical assumption, one that discrete collapse models [36] decline outright and that the algebraic formulation of field theory [30] has no sharp-time object to apply to, since it assigns observables to open regions and none to a

sharp time. The second is dynamical sufficiency of the value so identified, which indivisible stochastic dynamics [1] denies. Neither is delivered by the definition of a limit.

Observing a process through a temporal window of width δ replaces the state $X(t)$ by a functional of the path segment on $[t-\delta, t]$. This is a lumping of the path-segment process, not of the state, and the path-segment process of a Markov process is itself Markov; a lumping of a Markov chain is generically not Markov [7, 8], so the windowed family generically fails to compose through intermediate times for every $\delta > 0$. Whether the defect vanishes as $\delta \to 0$ depends on a further hypothesis. If the underlying process is an irreducible finite-state chain in its stationary law, observed through the window average of its state, the defect vanishes at least linearly in δ, which is Lemma 1 of Sec. 6; if the observation is instead a non-lumpable function of the state, the defect converges to that of the lumped family, which is generically positive, and when that limit is nonzero no refinement removes it. Under the first hypothesis, the limit point has a property, divisibility, that the members of the approximating family generically lack.

This argument applies within the standard continuum topology of the real line. Alternative topological structures, non-Hausdorff spacetimes in which limits need not be unique [9], or causal set theory in which the continuum is replaced by a discrete partial order [10], would modify the analysis in ways specific to each framework. The present argument targets the standard idealization.

The completeness of the state at a temporal point is therefore an additional assumption, not a consequence of how such states are constructed, and the following sections show that established physics gives positive reasons to doubt it.

3 Arguments from Relativity

General relativity and quantum mechanics are among the most precisely confirmed theories in the history of science. Applied jointly, they yield concrete bounds on the resolution of spatial and temporal measurement through three lines of argument, two for the spatial bound and one, from clock physics, for the temporal one. The bounds are obtained by extrapolating both theories toward a regime where neither is independently known to remain complete; they are robust across the approaches reviewed by Garay [12], but they are not theorems of a completed theory of quantum gravity. We distinguish throughout between a minimum measurable resolution, which the arguments establish, and stronger claims of spacetime discreteness or of the failure of point-indexed mathematical states, which they do not.

Mead [11] and Garay [12] establish that a fundamental minimal spatial length must exist as a consequence of the joint requirements of quantum mechanics and general relativity. The argument is forced by the uncertainty relations: through the de Broglie relation, improving spatial resolution to a scale δx requires concentrating momentum of order $\hbar/\delta x$, and therefore energy of order $\hbar c/\delta x$, within the probed region. At sufficiently small δx the Schwarzschild radius of this energy exceeds δx itself, and the region undergoes gravitational collapse. Since the fundamental length deduced by Mead scales as $\sqrt{G}$, the Planck length $l_P = \sqrt{\hbar G/c^3} \approx 1.6 \times 10^{-33}$ cm, the unique length constructible from $\hbar$, G, and c, is the natural lower bound, and both analyses yield results consistent with this scale, though neither derives it as a sharp numerical threshold. The existence of this fundamental length in turn limits the precision with which clocks can be synchronized, as Mead shows [11], an observation originally motivated by the problem of infinities in quantum field theory arising from point-particle idealizations, which connects the gravitational argument directly to the renormalization evidence of Sec. 4.

't Hooft's analysis of trans-Planckian scattering [13] arrives at the same spatial obstruction by a route that is arguably more direct: when two particles collide at sufficiently high center-

of-mass energy, general relativity requires that the interaction region undergo gravitational collapse. This conclusion depends only on the particle kinematics and the Einstein equations, without recourse to quantum interference effects, and therefore rests on fewer assumptions than the Mead-Garay derivation. The result concerns the scattering configuration; its bearing on temporal resolution is through the inference drawn below, not through any direct statement about pointwise descriptions.

A third route to a temporal bound is provided by Salecker and Wigner [14], who show that quantum mechanics constrains the precision of any clock used to measure spacetime intervals. An ideal clock measuring a time interval T with resolution δt must, to avoid quantum spreading of its center of mass degrading its accuracy, satisfy a minimum mass condition $M \gtrsim \hbar T/(c^2\,\delta t^2)$. This bound alone involves no gravitation. A Planck-scale value arises when it is combined with the requirement that the clock not collapse into a black hole, $r_s = 2GM/c^2 \lesssim \ell_{\text{clock}}$, together with the condition $\ell_{\text{clock}} \lesssim c\,\delta t$ that the clock be readable within its own resolution. The combination gives

$$\delta t \gtrsim \left(T\,t_P^2\right)^{1/3}, \qquad t_P = \frac{l_P}{c} = \sqrt{\frac{\hbar G}{c^5}} \approx 5.4 \times 10^{-44}\ \text{s}, \tag{2}$$

up to a factor of order unity, the form obtained by Károlyházy [15] and by Ng and van Dam [16], of which $\delta t \gtrsim t_P$ is the corollary for $T \sim \delta t$. The dependence on T is disputed: Diósi and Lukács [17] argue that the bound overstates the uncertainty of spacetime, and Baez and Olson [18] show that a clock attached to a massive body reduces the uncertainty to the Planck scale itself. Only the Planck-scale corollary is used here. The route is independent of the Mead and 't Hooft analyses in that it constrains the clock rather than the probed region, not in that it dispenses with gravitation.

The temporal consequence of the spatial results follows independently, by an inference that uses only the position-momentum relation. A signal that resolves a time interval δt is localized in time to δt and therefore in space to $c\,\delta t$, and localization to $\delta x = c\,\delta t$ requires momentum of order $\hbar/\delta x$ and hence energy of order $\hbar/\delta t$. Since the Planck time t_P is the light-crossing time of the Planck length, probing temporal intervals shorter than t_P requires concentrating energy at the same densities that trigger gravitational self-collapse in the Mead and 't Hooft analyses; the energy scales are identical, related by the factor c. This temporal extension is not stated by Mead, Garay, or 't Hooft, but follows from their spatial results by this route, which does not invoke the time-energy uncertainty relation. The Mead and 't Hooft analyses, together with this inference, imply that the interaction region collapses into a black hole before the temporal measurement completes. The Salecker-Wigner route arrives at the same temporal bound from a different direction: the quantum-mechanical properties of the clock, combined with the requirement that the clock itself not collapse, prevent arbitrarily fine temporal resolution without any appeal to the collapse of the measured region.

These bounds follow from the two established theories themselves rather than from the modeling assumptions of any particular approach to quantum gravity, and they do not rely on the time-energy uncertainty relation, whose interpretation is complicated by the fact that time is not an observable in standard quantum mechanics. The specific physics below t_P is unknown; within the semiclassical domain in which the bounds are derived, they obstruct the limiting process $\delta t \to 0$ for the protocols they describe, a probe localized by concentrated energy and a clock read within its own resolution.

This is an operational result. It establishes an obstruction to reaching the point within that domain; it does not by itself establish that the mathematical state assigned to the point is incomplete, and no such claim is made here. Its bearing on completeness is that the direct route by which the completeness of a point-indexed state could be confronted with experiment, the progressive refinement of temporal resolution, is closed at a finite scale. Completeness at

the point is therefore an assumption that no measurement within that domain can underwrite, and Sec. 2 has shown that it is not underwritten by the construction either.

Approaches to quantum gravity that begin with a discrete structure at the Planck scale, lattice gravity, causal set theory [10], and spin foams, are compatible with this conclusion: they abandon the continuum temporal point from the outset, replacing the standard idealization with a fundamentally different mathematical structure in which the instantaneous state, as conventionally conceived, plays no role.

The preceding section established that the mathematical construction of the point-state does not supply dynamical completeness at the point. This section establishes an obstruction to the physical process required to reach the point: within the semiclassical domain of the arguments, gravitational self-collapse and quantum clock bounds independently prevent the limiting process from being completed.

4 Argument from Renormalization

Quantum field theory provides corroboration from an independent direction. Quantum fields are operator-valued distributions, and products of field operators at coincident spacetime points are singular. The ultraviolet divergences of perturbation theory are the perturbative expression of this fact: they arise whenever the formalism is pushed to the point-valued limit [19].

The renormalization program [20, 21, 22, 23, 24, 25], given conceptual foundation by Wilson [23, 24, 25], extracts finite physical predictions by introducing a regulator, a momentum cutoff Λ_0 or a lattice spacing, that keeps the theory away from the spacetime point, and by expressing predictions through renormalized quantities from which the regulator can then be removed. Four scales enter this procedure: the regulator; the renormalization scale μ at which renormalized parameters are defined, on which exact observables do not depend, that independence being the content of the renormalization-group equation; the validity scale Λ of an effective description, beyond which new degrees of freedom enter and the expansion of predictions in powers of E/Λ ceases to hold; and the resolution of any particular experiment. The measured running of the strong coupling with μ [28] confirms the renormalization-group structure but is not itself a limit on resolution, since the measured quantity is a scheme-dependent parameter at a chosen scale. The argument of this section rests on the validity scale Λ and on the coincident-point singularity, not on μ.

Where a continuum limit exists, it is obtained by removing the regulator from renormalized quantities, not by evaluating the bare theory at a point, and whether an interacting continuum limit exists at all is theory-dependent. For lattice-regularized scalar ϕ^4 theory in four dimensions, the scaling limit at criticality is now known to be Gaussian, that is, trivial [27]: removing the regulator removes the interaction. Asymptotically free theories, in which the flow reaches the Gaussian fixed point in the ultraviolet, are one route to an interacting continuum limit; an interacting ultraviolet fixed point is another. In every case, the predictions that agree with experiment to extraordinary precision [29] are computed from renormalized quantities; where a regulator is removed, it is removed from those quantities, and the bare theory at a point is never evaluated.

The Wilsonian understanding of renormalization differs from the earlier view in which divergences were regarded as computational artifacts of perturbative methods: Wilson's framework demonstrates that the need for renormalization is not an artifact of perturbation theory. In Wilson's picture, the description at cutoff Λ_0 is an effective one, obtained by integrating out the degrees of freedom above Λ_0, with parameters that flow as Λ_0 is lowered; a continuum limit, $\Lambda_0 \to \infty$, requires an ultraviolet fixed point of that flow, and the lattice at fixed spacing is

itself an effective description of this kind [26]. On a lattice of fixed spacing all quantities are finite; renormalization enters in relating descriptions across scales and in taking the continuum limit. The bare theory at a spacetime point is the $\Lambda_0 \to \infty$ end of the flow, not an object whose properties the effective description fixes, and its status, where a continuum limit exists at all, is settled theory by theory, as the triviality of ϕ^4 in four dimensions illustrates. The renormalization group is the systematic expression of this fact.

The formalism becomes singular at a spacetime point, and the physical content of an effective description is organized by its validity scale. Whether a future theory could render the spacetime point well defined is an open question; the argument here is evidential, not deductive. Our best established theories, whose predictions have been confirmed to extraordinary precision, provide no basis for assuming that the point carries complete physical information, and positive grounds for doubting it.

The algebraic formulation makes the localization of physical content precise in two respects, which bear on this paper in opposite directions. The first concerns observables. The Haag-Kastler axioms assign observable algebras to open spacetime regions [30], and Wightman's theorem forecloses bounded fields at points: a function B from spacetime to bounded operators, with $B(x)$ in the algebra of every region containing x, translation covariant and commuting at spacelike separation, in a theory satisfying the spectrum condition with a unique vacuum, is a multiple of the identity, $B(x) = b\,\mathbf{1}$ for every x [33], which is Wightman's ground for taking fields to be operator-valued distributions. The second concerns the data that fix the dynamics: the time-slice axiom, or primitive causality [31, 30], states that the algebra of any neighborhood of a Cauchy surface, however thin, is already the algebra of the whole spacetime. It holds for free fields, where it follows from the field equation, and for perturbatively renormalized scalar theories [32]. The dynamics is thus determined by a slab of nonzero thickness, however small, which is an initial-value structure; the axioms assign no algebra to the sharp surface itself. The framework therefore assigns physical content to thin extended regions and has no object at a sharp time, which is the structure this paper argues for. These results concern the localization of observables and of the data that fix the dynamics; they do not by themselves bear on the composition of two-time kernels, which in that framework is carried by the automorphism group. The parallel with the preceding arguments is structural: in all three cases the physics fails at the temporal or spacetime point, and the content lives in an extended region.

5 The Collapse-Timing Formulation of the Measurement Problem

The measurement problem is standardly formulated as the tension between the deterministic, unitary evolution of the quantum state and the occurrence of definite, stochastic outcomes upon measurement [35]. One formulation of this tension presupposes that both sides of the measurement event, the unitarily evolving superposition before and the definite outcome after, are complete states at temporal points, where "complete" means carrying sufficient dynamical information to determine future evolution (whether pure or mixed). The demand is that the timeline can be cut at "the moment of collapse" with a fully specified state on either side. This is precisely the demand that the instantaneous state carry complete dynamical information, applied to the measurement event. We call this the collapse-timing formulation.

Measurement is a physical interaction between system and apparatus. Every stage, photons hitting detectors, electrons cascading through amplifiers, pointers moving on dials, takes nonzero time. There is no operationally privileged instant at which the measurement happens; system and apparatus become correlated over an interval, a picture independently supported by decoherence theory, which models the finite timescale over which system-environment entan-

glement suppresses interference [34, 35]. At the beginning, they are uncorrelated. At the end, the apparatus registers a definite outcome.

Asking "at which point in between did the transition occur?" is demanding dynamical completeness at intermediate temporal points within the measurement interval. The preceding arguments show this demand is unjustified: the mathematical construction does not supply completeness at the point, the physical limit to the point is obstructed within the domain of established physics, and known physics becomes singular at the point-valued limit.

The collapse-timing formulation of the measurement problem is thereby dissolved rather than solved. The question of the instant at which the transition occurs presupposes complete descriptions at temporal points on either side; without that presupposition the collapse-timing question has no well-defined terms in which to arise. The conclusion is that no operationally privileged instant exists. It is not the claim that no theoretical jump time can exist, which the discrete collapse models cited in Sec. 2 posit and which the present arguments do not exclude. The broader tension between unitary evolution and definite outcomes does not depend on an instant, as the continuous stochastic dynamics of spontaneous localization models make explicit [37, 38], and other aspects of the measurement problem, the derivation of the Born rule, the definite-outcome problem, and the preferred-basis problem, are not addressed by this dissolution and require independent treatment.

The specific outcome of a measurement, why *this* result rather than *that* one, is not addressed by this dissolution. Within the framework of indivisible stochastic processes [1, 2, 3], established for systems with finite configuration spaces and unistochastic transition matrices, the outcome is an irreducibly stochastic draw from the transition matrix over the measurement interval. In the standard Hilbert space formulation, the Schrödinger equation is deterministic, making stochastic outcomes appear to require a mechanism, collapse, that violates the governing equation. In the indivisible stochastic formulation, the dynamics are stochastic from the start; the transition matrix yields probabilities and outcomes are draws from those probabilities, with no deterministic equation being violated. As Barandes establishes [1, 2], the Schrödinger equation deterministically governs the evolution of the transition matrix, the probability law itself, not the realization of individual outcomes, which was never deterministic in either formulation. Within the scope of the correspondence the two formulations are empirically equivalent: they produce identical predictions. Within that framework, then, the conceptual puzzle is an artifact of the Hilbert-space formulation's mathematical structure rather than of the physics; this is a conclusion inside the stochastic formulation, not a consequence of standard quantum theory.

6 Consequences

If completeness at a temporal point is an assumption rather than a consequence, then every description that presupposes it is an idealization whose adequacy must be established for the system at hand. In the stochastic setting that description is the Markov property, Definition 1(b), and its observable consequence is the Chapman-Kolmogorov composition of the two-time kernels through intermediate times,

$$\Gamma(t_2 \leftarrow t_1) = \Gamma(t_2 \leftarrow t)\,\Gamma(t \leftarrow t_1), \qquad t_1 < t < t_2. \tag{3}$$

By the hierarchy stated in Sec. 1, a failure of (3) refutes the Markov property on the state space in question, while its holding does not establish it.

First, as noted in Sec. 2, a process observed through a temporal window of finite width is generically non-Markovian on the observed values [7, 8], and whether the composition defect disappears under refinement depends on whether the underlying process is Markov on the observed state space.

Lemma 1. *Let X be an irreducible continuous-time Markov chain on a finite set $S \subset \mathbb{R}$, started in its stationary law π, with generator Q and maximal jump rate $q_* = \max_x |Q_{xx}|$. For $\delta > 0$ let $Y_\delta(t)$ be the average of X over $[t-\delta, t]$; fix $t_1 < t < t_2$ with $t - t_1 > \delta$ and $t_2 - t > \delta$; let $V_\delta(u \leftarrow s)$ denote the two-time kernels of Y_δ, the regular conditional distributions of $Y_\delta(u)$ given $Y_\delta(s)$; and let $\Delta_\delta = V_\delta(t_2 \leftarrow t_1) - V_\delta(t_2 \leftarrow t)\, V_\delta(t \leftarrow t_1)$ be the composition defect. Then there is a constant C, depending only on Q and on $t - t_1$, such that the signed measure $\Delta_\delta(\,\cdot \mid y)$ has total variation at most $C\, q_*\, \delta$ for almost every y with respect to the law of $Y_\delta(t_1)$, for all $\delta \le \min\{(t-t_1)/2,\, 1/(2q_*)\}$. If instead the observed quantity is the window average of $\lambda(X)$ for a map $\lambda\colon S \to S'$ under which the lumped process $\lambda(X)$ is not Markov, then as $\delta \to 0$ the two-time kernels of the observed process converge to those of $\lambda(X)$, and the defect converges to that of the lumped family, which is generically positive and independent of δ.*

Proof. Let E be the event that X has no jump in $[t-\delta, t]$. Holding times are exponential with rates at most q_*, so $\mathbb{P}(E^c \mid \mathcal{F}_{t-\delta}) \le 1 - e^{-q_*\delta} \le q_*\delta$, and on E one has $Y_\delta(t) = X(t)$. Write $\mu_y(x) = \mathbb{P}(X(t_1) = x \mid Y_\delta(t_1) = y)$ and $\nu_{y'}(z) = \mathbb{P}(X(t) = z \mid Y_\delta(t) = y')$ for regular conditional distributions, defined for almost every y and y' with respect to the laws of $Y_\delta(t_1)$ and $Y_\delta(t)$, and $K(B \mid z) = \mathbb{P}(Y_\delta(t_2) \in B \mid X(t) = z)$. Since $Y_\delta(t_1)$ is measurable with respect to the past of t_1, and the windows defining $Y_\delta(t)$ and $Y_\delta(t_2)$ lie after t_1 and after t respectively, the Markov property of X at t_1 and at t gives

$$
\begin{aligned}
V_\delta(t_2 \leftarrow t_1)(B \mid y) &= \sum_x \mu_y(x) \sum_z \mathbb{P}(X(t) = z \mid X(t_1) = x)\, K(B \mid z),\\
V_\delta(t \leftarrow t_1)(y' \mid y) &= \sum_x \mu_y(x)\, \mathbb{P}(Y_\delta(t) = y' \mid X(t_1) = x),\\
V_\delta(t_2 \leftarrow t)(B \mid y') &= \sum_z \nu_{y'}(z)\, K(B \mid z),
\end{aligned}
$$

where sums over y' are to be read as integrals wherever $Y_\delta(t)$ has a continuous part. Hence $\Delta_\delta(B \mid y) = \sum_x \mu_y(x) \sum_z K(B \mid z)\, D_x(z)$ with

$$
D_x(z) = \sum_{y'} \Big[\mathbb{P}\big(X(t) = z \mid Y_\delta(t) = y',\, X(t_1) = x\big) - \nu_{y'}(z) \Big]\, \mathbb{P}\big(Y_\delta(t) = y' \mid X(t_1) = x\big),
$$

and since each $K(\,\cdot \mid z)$ is a probability measure and μ_y is a probability measure on S, the total variation of $\Delta_\delta(\,\cdot \mid y)$ is at most $\max_x \sum_z |D_x(z)|$ for almost every y, the bound being uniform in x. Split the sum over y' according to whether $y' \in S$. Values $y' \notin S$ occur only on E^c, so their total weight is at most $q_*\delta$, and each bracket has ℓ^1 norm at most 2. For $y' \in S$, let $c = \min\{\min_x \pi(x),\ \min_{x,z} \min_{s \in [(t-t_1)/2,\, t-t_1]} (e^{sQ})_{xz}\}$, which is positive by irreducibility. Each conditioning event in the bracket has probability at least $c\, e^{-q_*\delta} \ge c/2$, because $\{Y_\delta(t) = y'\} \supseteq \{X(t-\delta) = y'\} \cap E$, and $X(t) = y'$ on E. Both conditional laws of $X(t)$ in the bracket therefore assign mass at least $1 - 2q_*\delta/c$ to the point y', so their ℓ^1 distance is at most $8q_*\delta/c$. Altogether $\sum_z |D_x(z)| \le (2 + 8/c)\, q_*\delta$, which is the claim with $C = 2 + 8/c$. For the second statement, $Y_\delta(t) \to \lambda(X(t))$ almost surely as $\delta \to 0$ at each fixed time, since almost surely no jump occurs at that time; the joint laws at (t_1, t, t_2) converge, the kernels converge on values of positive limiting mass, and the limit family is that of $\lambda(X)$, whose defect is generically positive when the lumping is not Markov [8]. □

The memoryless description is therefore a property of the idealized limit that the finite-resolution descriptions of the same process generically lack, and Sec. 3 shows that the limit is obstructed within the domain of established physics. The lemma yields a one-sided test. Under its hypotheses, the composition defect of the window-averaged family must fall to zero at least linearly as the window is refined; so for a finite-state system observed in its stationary law,

a defect that persists under refinement is memory intrinsic to the chosen state space rather than to the resolution, and refutes the memoryless description on that space. A defect that vanishes is consistent with the memoryless description but does not establish it, since non-Markov processes with composing two-time kernels exist [4]. The scaling of the defect with resolution thus separates two sources of Chapman-Kolmogorov failure, and a persistent defect carries the evidential weight. The lemma bounds the defect from above and supplies no lower bound, so the defect of a given system may vanish faster than linearly or vanish identically. An application of the test must therefore declare a sampling floor, the window width below which the estimated defect is not distinguishable from sampling error at the sample size in hand, calibrated for the model in question by simulation or bootstrap rather than derived from the lemma; the test is silent below the floor, and its evidence is a defect that persists above it under refinement.

Second, Barandes's stochastic-quantum correspondence [1, 2, 3] establishes that quantum systems, described on their configuration spaces, are indivisible stochastic processes whose transition matrices compose through intermediate times only at division events, at any resolution; the correspondence is established for finite configuration spaces and unistochastic transition matrices, and the claims here are confined to that scope. The burden of justification therefore lies with the memoryless description, whether the system is classical or quantum.

Distinguishing state-space memory of classical origin from quantum indivisibility requires a criterion beyond the composition defect and its scaling, since classical hidden variables also produce a resolution-independent defect. The criterion is consistency under intermediate observation. A classical process observed non-invasively satisfies the Kolmogorov consistency conditions: its statistics at two times are recovered by marginalizing its statistics at three times over the unobserved intermediate value. A quantum process on its configuration space generically violates them, since an intermediate measurement whose outcome is discarded alters the later statistics, the two-slit experiment being the elementary case, and the process-tensor framework isolates this violation rather than the composition defect [41]. The criterion inherits the non-invasiveness assumption familiar from tests of macrorealism [42]. Its quantitative development, together with the divisibility measures for open quantum systems [39, 40], lies outside the scope of this paper.

7 Discussion

The three arguments assembled here draw only on established mathematics and physics, and none invokes a particular interpretation of quantum mechanics, proposes new axioms, or modifies existing theory. The argument from limits requires only the ε-δ definition of a mathematical limit and the elementary fact that a lumping of a Markov process is generically not Markov. The argument from relativity requires only quantum mechanics and general relativity, each in its established domain and jointly extrapolated to the Planck scale, where, within the semiclassical domain of that extrapolation, gravitational self-collapse and Salecker-Wigner clock bounds independently obstruct the reaching of the temporal point. The argument from renormalization requires only the existence of ultraviolet divergences and their physical interpretation through Wilson's renormalization group. The new element is the joint use of these results: together they remove the basis for treating the completeness of the instantaneous state as given, without invoking any physics beyond these established theories, although, as Sec. 3 states, the Planck-scale bounds extrapolate two of them beyond their tested domains.

The nearest neighboring discussion lies in the philosophy of mechanics. Albert [43] requires that an instantaneous state be logically independent of the states at other instants, concludes that the instantaneous state of a Newtonian system consists of positions alone, since a velocity

is a rate of change of position, and so obtains a state that describes the instant completely and does not fix the motion. Butterfield [44], following Arntzenius [45], argues against what he calls pointillisme, the doctrine that a theory's fundamental quantities are intrinsic to spacetime points, on the ground that instantaneous velocity is fixed by the trajectory in arbitrarily small neighborhoods of an instant and not by the instant alone. The present claim differs in object and in evidence: it concerns the dynamical sufficiency of the state rather than the instantaneity or intrinsicness of a quantity, and its grounds are the Planck-scale bounds and the field-theoretic localization of content rather than the analysis of velocity. The three are compatible; the present arguments reach the anti-pointilliste conclusion by a physical route.

The conclusion argued here is a reversal of the burden of proof about an assumption, not about the metaphysics of the instant. The completeness of the instantaneous state is not delivered by the construction that defines it, is not reachable by measurement within the domain of established physics, and is not where established theories locate their physical content. It is an assumption, and the descriptions that presuppose it, the initial-value formulation, the Markov property, and the instantaneous collapse of the quantum state, are idealizations whose warrant must come from elsewhere, with the composition of two-time kernels as the observable through which the Markov property is tested. That foundational claim is a claim about warrant and is not itself an empirical prediction; the empirical content is stated in Secs. 3 and 6: a measurement resolving time intervals below t_P would refute the Planck-scale bounds, and the refinement behavior of the composition defect, Lemma 1, is the observation-model prediction, falsifiable in the direction stated there. Once completeness is no longer presupposed, the memoryless description is the special case that must be justified, and the dissolution of the collapse-timing measurement problem follows as one instance of that reversal rather than as a separate result. In practice, memorylessness is not to be presumed without a test of its stability under refinement of the temporal resolution and under changes of the state-space specification; Sec. 6 supplies the test, and the two halves of Lemma 1 are those two changes. The test is relative to the observed state space, and a formulation of incompleteness that is independent of the description is not given here: augmenting the state with its full history renders any process Markov, so such a formulation requires a physically fixed class of admissible descriptions, and this paper fixes that class only in part, by the resolution bound of Sec. 3 and the non-invasiveness condition of Sec. 6. Neither constraint bounds history depth or state-space dimension: a history state on a time grid coarser than the resolution bound satisfies both conditions and has zero defect, so the class is unpinned where augmentation acts.

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